Slopes of lines

1 Define the slope of a line.

2 What is the change in the vertical distance along a line called?

3 What is the change in the horizontal distance along a line called?

4 Explain what it means if the slope of a line is zero.

5 Explain what it means if the slope of a line is undefined.

6 If we have two points and the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)? Explain.

7 Using $m = \dfrac{\text{rise}}{\text{run}}$, find the slope of the line that passes through:

a Point A $(-1, 0)$ and Point B $(0, 3)$

b Point A $(2, -6)$ and the origin

8 Using $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, find the slope of the line that passes through:

a Point A $(3, 5)$ and Point B $(1, 8)$

b Point A $(-3, -1)$ and Point B $(-5, 1)$

9 Find the slope of the line that passes through the points:

a $(3, 4)$ and $(5, 4)$

b $(-3, 4)$ and $(1, 4)$.

10 Consider the points $A\,(24, -56)$, $B\,(-6, -2)$ and $C\,(-11, 7)$:

a Find the slope of AB.

b Find the slope of BC.

c Do the points A, B and C lie on a single straight line?

11 A line passing through the points $(5, 3)$ and $(2, t)$ has a slope equal to -4.
Find the value of t.

12 A line passes through the points $(11, c)$ and [illegible] $16)$ and has a slope of $-\frac{4}{7}$.

Find the value of c.

13 The distance between points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

This formula has been used as shown: $d = \sqrt{(3-6)^2 + (7-4)^2}$. Determine the slope of the line through AB.

14 What is the slope of the segment joining Point A $(1, -1)$ and Point B $(-1, -2)$?

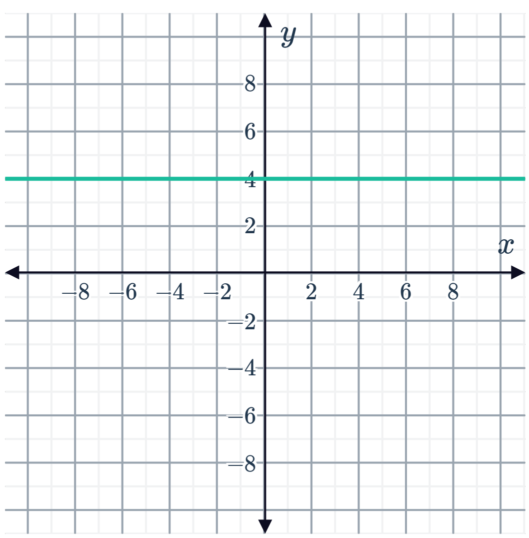
15 The slope of a line is 7 and A $(9, 7)$ is a point on the line. Another point, B, on the line has $y = 56$. What is the x-value at this point, as denoted by d?

16 For each of following pairs of coordinates, determine whether it will have an undefined or defined slope .

- **a** $(-10, 5)$ and $(-10, 12)$
- **b** $(10, 5)$ and $(10, 1)$
- **c** $(10, -1)$ and $(-10, -1)$
- **d** $(-10, 5)$ and $(10, 5)$
- **e** $(10, 7)$ and $(10, 2)$
- **f** $(10, -2)$ and $(-10, -2)$
- **g** $(-10, 7)$ and $(-10, 12)$
- **h** $(-10, 7)$ and $(10, 7)$

17 Examine the graph and determine the slope of the line.

a



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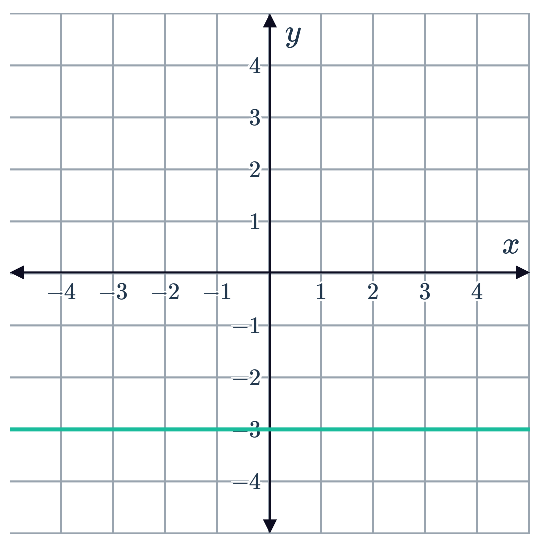
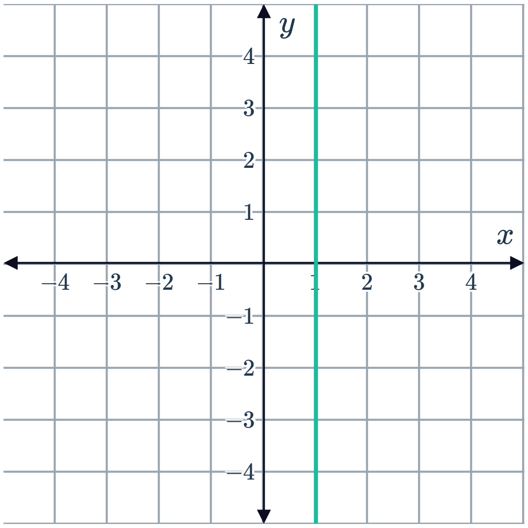
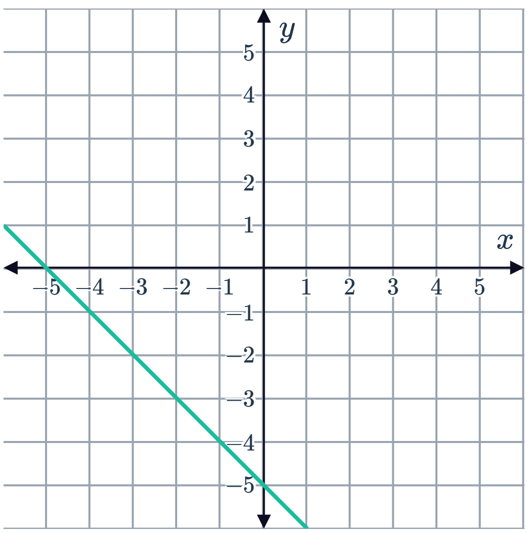


C

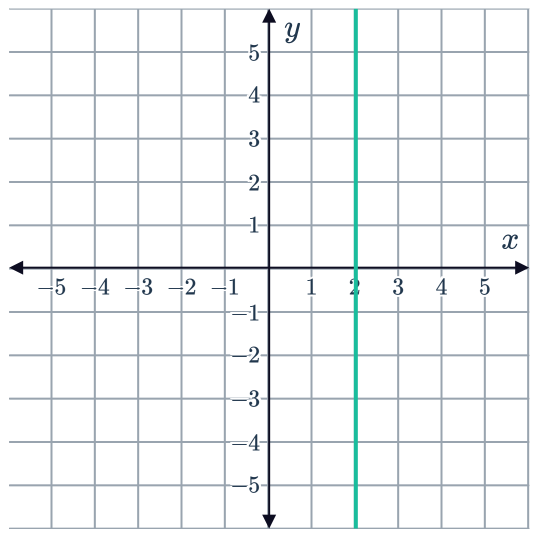


18 Consider the following graphs.

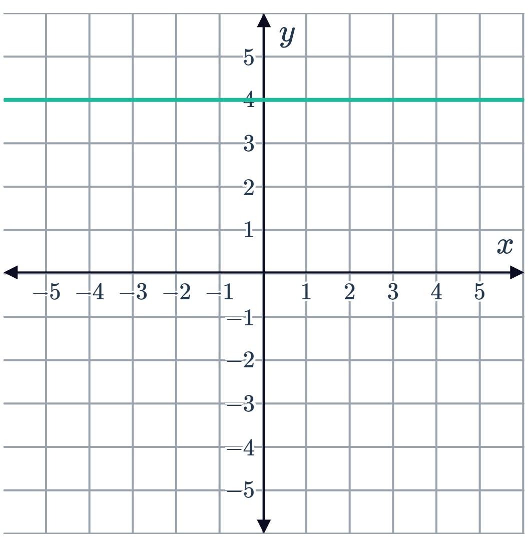
A



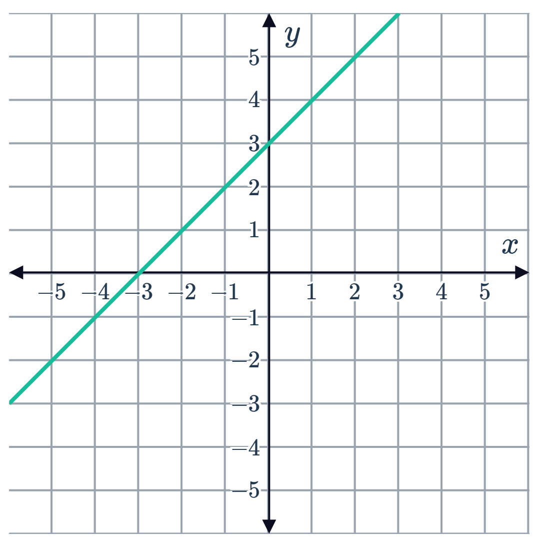
B



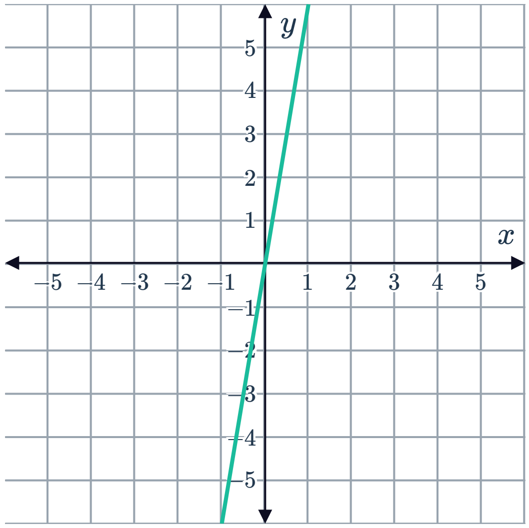
C



D



E



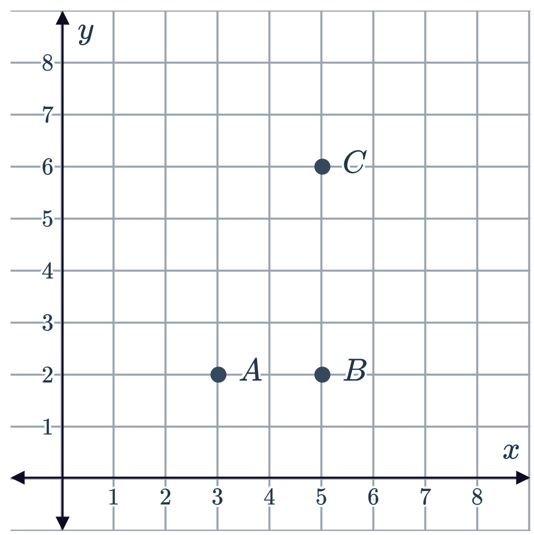
a Which line has a slope of 0?

b Which line has a slope that is undefined?

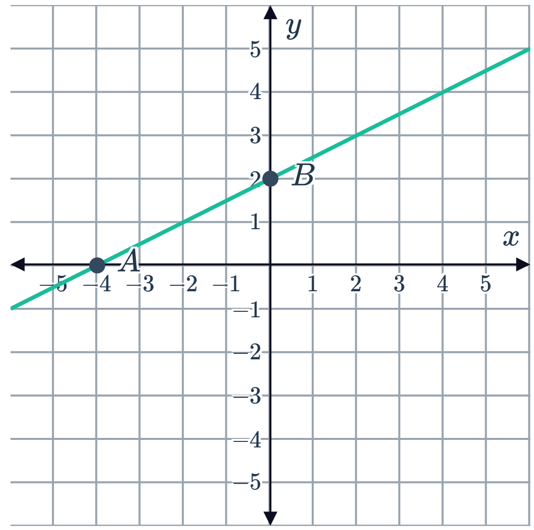
19 Consider the points on the graph.

a What is the slope of the line segment AB?

b What is the slope of the line segment BC?



20 Consider the segment shown in the graph w[redacted]oint A $(-4,0)$ and Point B $(0,2)$

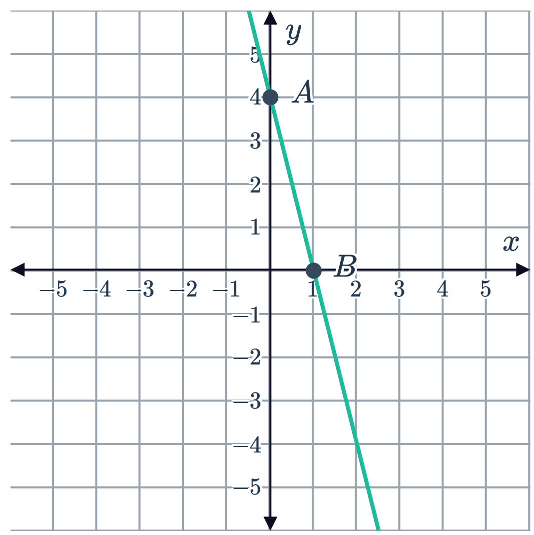


a Find the rise (change in the y value) between point A and B. Note: Ensure you have the correct sign.

b Find the run (change in the x value) between point A and B. Note: Ensure you have the correct sign.

c Find the slope of the segmentAB.

21 Consider the segment containing Point $A\,(0,4)$ and Point $B\,(1,0)$



a Find the rise (change in the y-value) going from point A to point B.

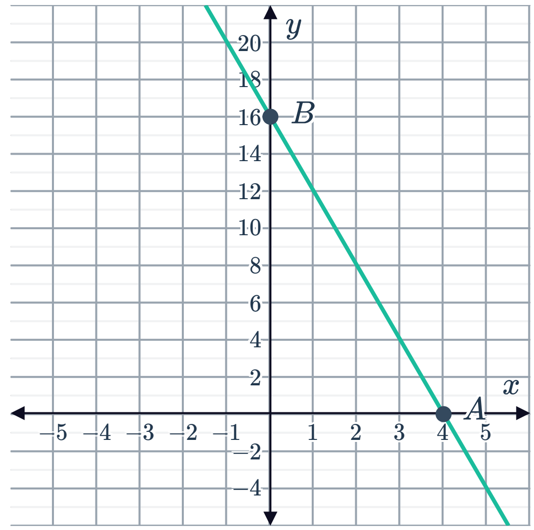
b Find the run (change in x-value) going from point A to point B.

c Find the slope of the segment AB.

22 Consider the line below, where Point A $(4,$ [redacted] Point B $(0, 16)$ both lie on the line:

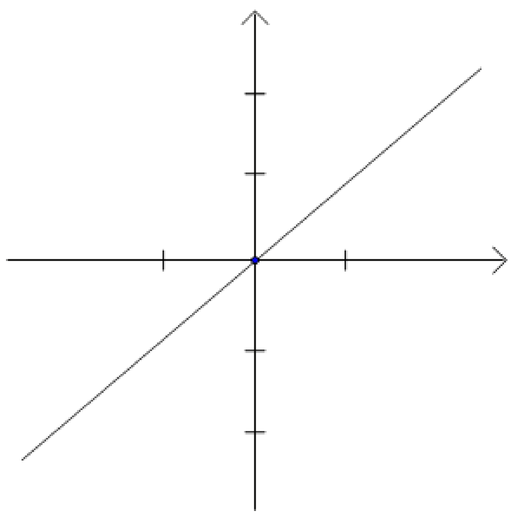
a What is the slope of the line?

b As x increases, is the value of y increasing or decreasing?

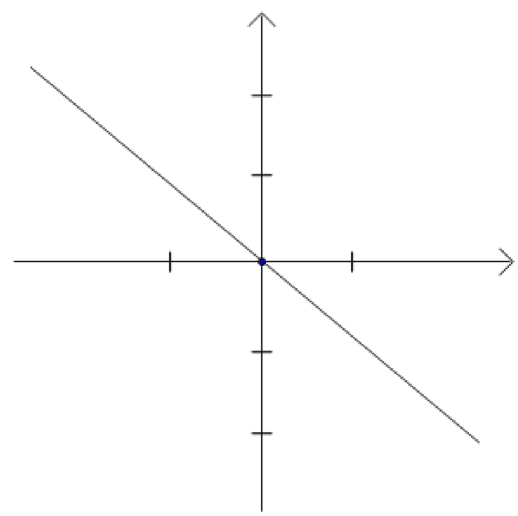


23 What kind of slope does each of the following lines have?

a

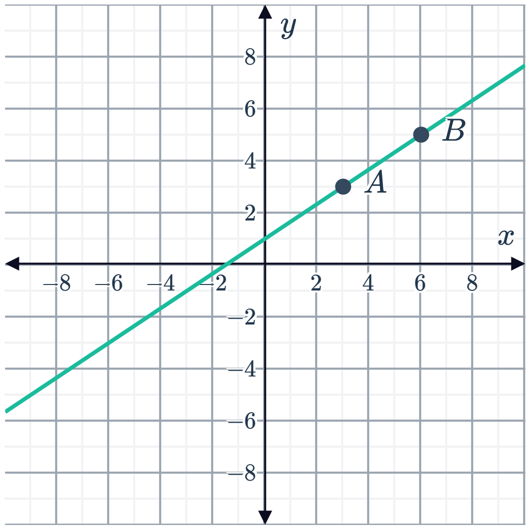


b

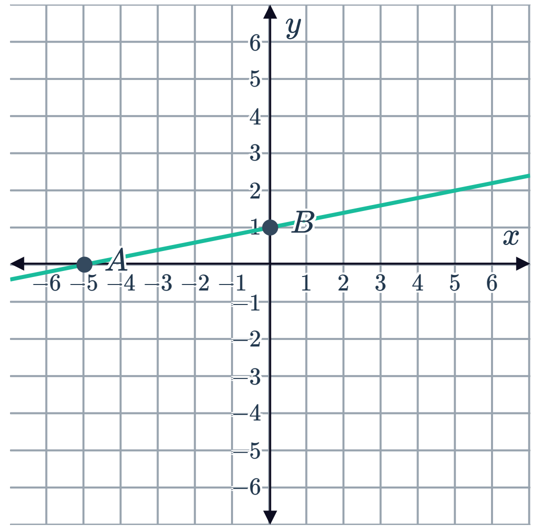


24 What is the slope of the line shown in the g[redacted]given that:

a Point A $(3, 3)$ and Point B $(6, 5)$ both lie on the line.

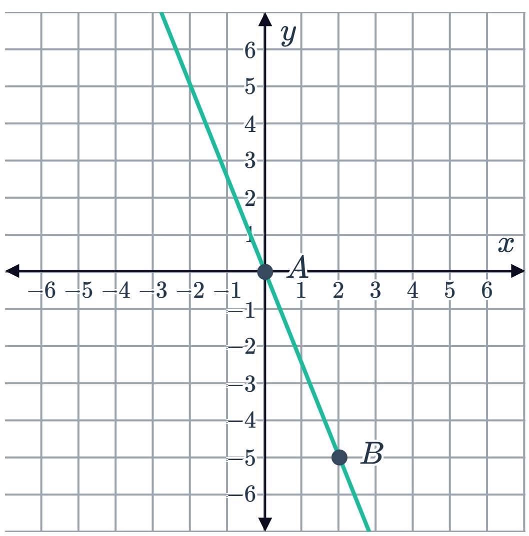


b Point A $(-5, 0)$ and Point B $(0, 1)$ both lie on the line.

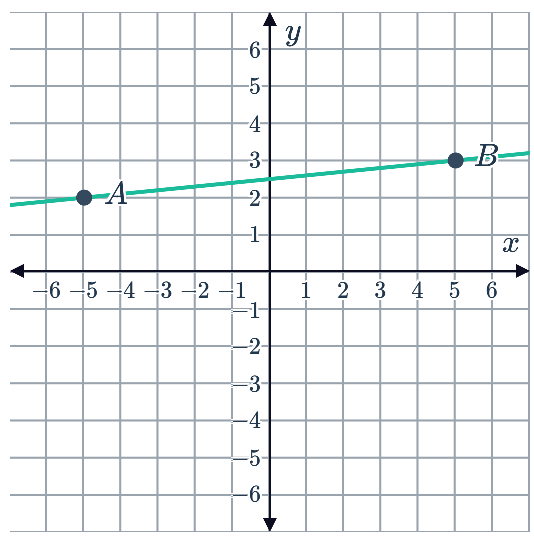


25 What is the slope of the line going through A and B?

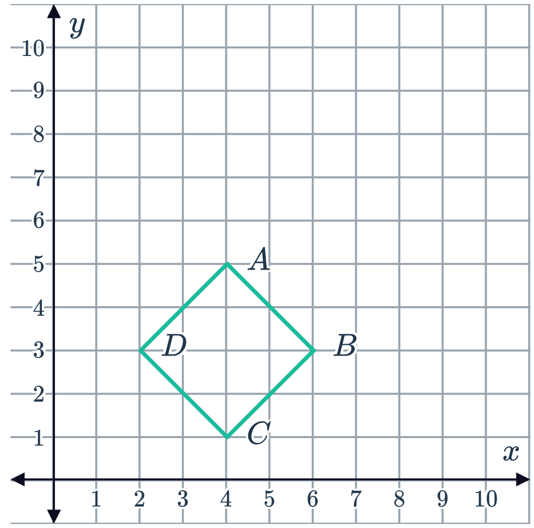
a



b



26 The four vertices of square ABCD have been plotted on a number plane.

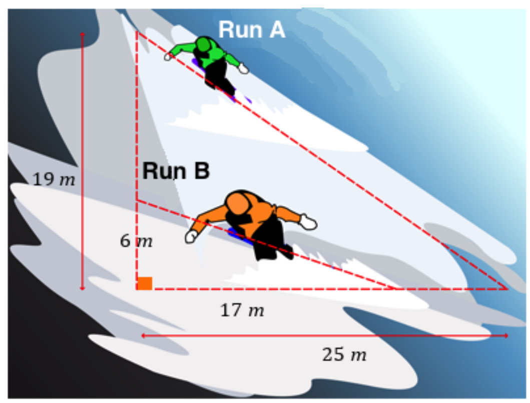


a Find the slope of side AB.

b Find the slope of side BC.

c What is the product of the slope s in part (a) and (b)?

27 A certain ski resort has two ski runs.



a Find the slope of ski slope A. Give your answer as a decimal to two decimal places.

b Find the slope of ski run B. Give your answer as a decimal to two decimal places.

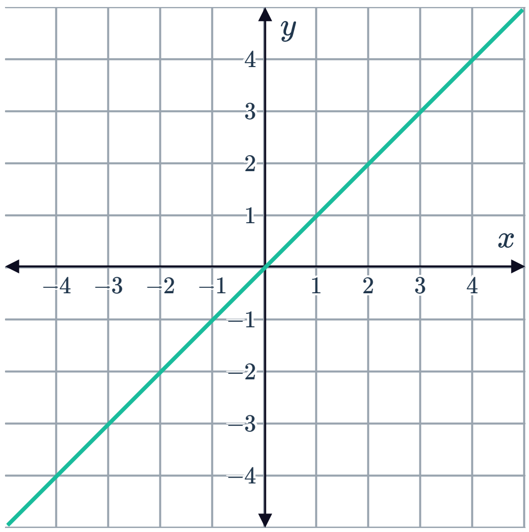
c Which run is steeper, A or B?

28 A staircase is to be built so that its maximum steepness is 0.6. If each step goes in 22 cm, what is the maximum height in centimeters it can rise vertically?

Additional questions

29 For each of the following lines, identify the slope .

a



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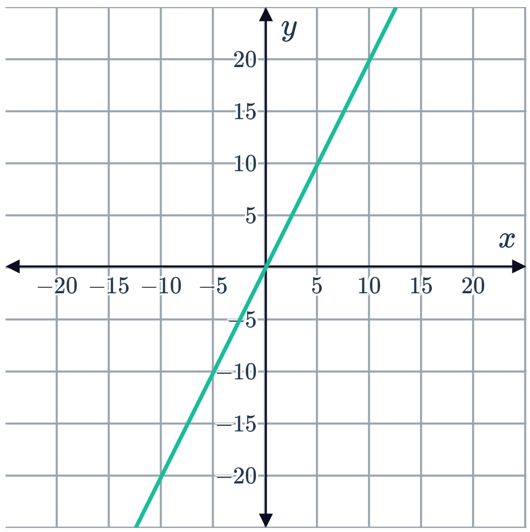
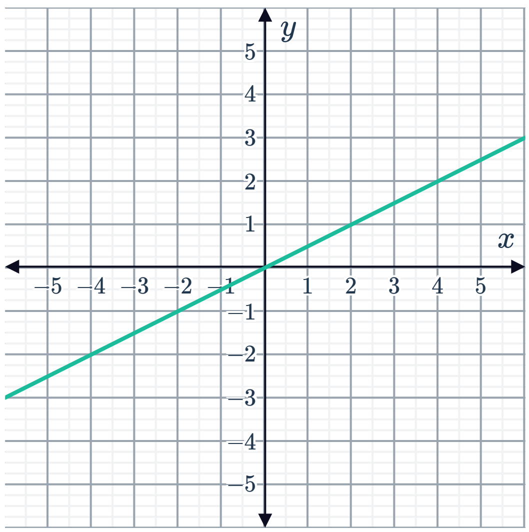
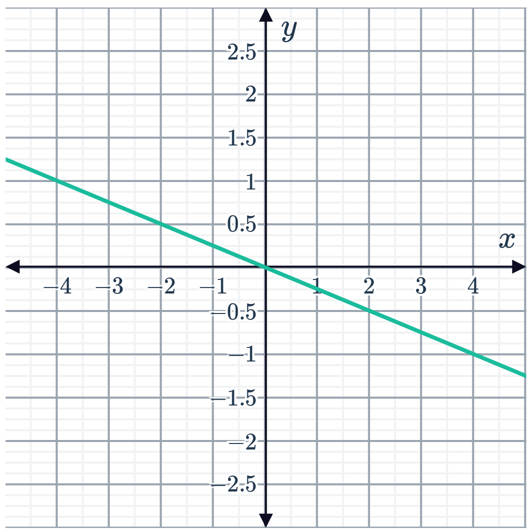


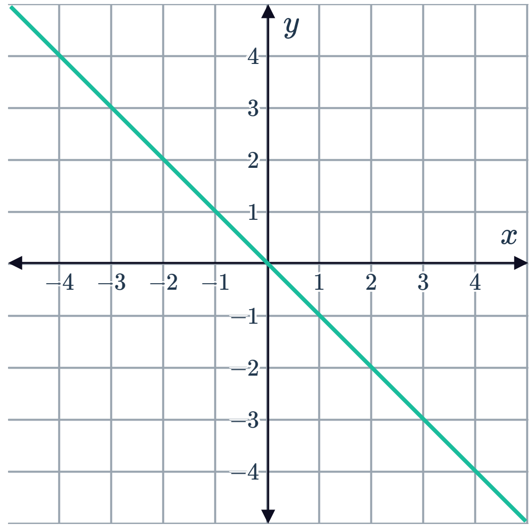
c

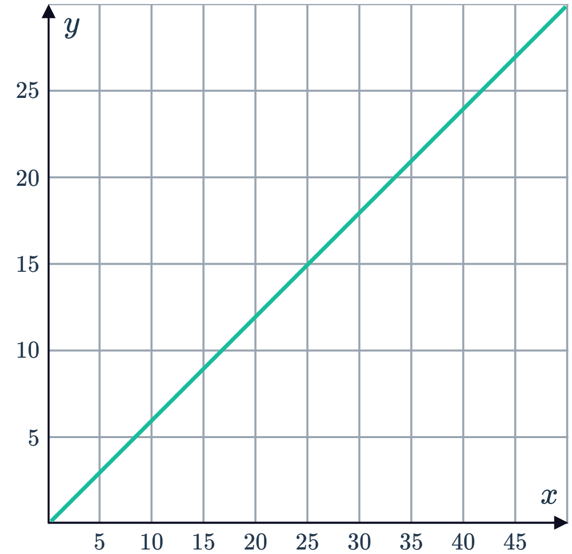


d



e





30 For each of the following relationships, identify the slope :

a

x	y
-3	-27
-2	-18
-1	-9
0	0
1	9
2	18

b

x	y
-8	-24
-6	-18
-4	-12
-2	-6
0	0
2	6

c

x	y
0.2	1.5
0.4	3
0.6	4.5
0.8	6
1	7.5
1.2	9

d

x	y
5	-7
10	-14
15	-21
20	-28
25	-35
30	-42